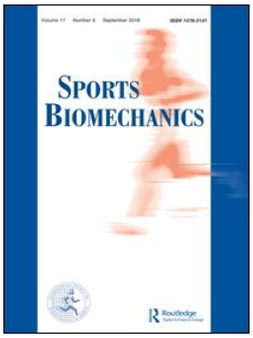

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Assisted or unassisted Nordic Hamstring Exercise? — Resistance exercise determinants at a glance

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ABSTRACT

The Nordic Hamstring Exercise (NHE) effectively strengthens the knee flexors. Typically conducted without assistance, extended knee angles are not reached with sustained muscle activation in the presence of insufficient eccentric strength and/or fatigue. This might impair the desired neuromuscular adaptations and assessment accuracy. This study investigated kinetic and kinematic differences between assisted and unassisted NHEs (3 × 3 repetitions) performed by sixteen male sprinters (22 years, 181 cm, 76 kg). Kinetic (peak moment, impulse) and kinematic parameters (e.g., time under tension, range of motion to excessive downward acceleration (ROM_{DWA}) were investigated. All analysed parameters significantly differed between assisted and unassisted NHEs ($p \leq 0.003$; $0.635 \leq \eta p^2 \leq 0.929$) favouring assisted execution, except for peak moments and maximal hip flexion. Repetition 1 of assisted NHEs revealed 21% higher impulses rising to 82% during repetition 9. Equivalent interactions of mode and repetition became apparent for time under tension, ROM_{DWA}, mean and fractional angular velocity. Unassisted NHEs elicited substantially greater inter-repetition fatigue (rep1 vs. rep9): +79% fractional angular velocity (d = 1.01), −41% impulse (d = 1.53), −31% ROM_{DWA} (d = 0.99) and −29% time under tension (d = 1.45). Assisted NHEs ensured higher execution quality and lower between-participant variability by facilitating a controlled full-ROM movement. Three sets of 3 NHEs sufficed to induce substantial fatigue within and across sets.

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Eccentric resistance training; exercise quality; hamstring strength; kinematic analysis; time under tension

Introduction

The Nordic Hamstring Exercise (NHE) is an effective resistance training exercise to improve knee flexor strength, thigh muscle balance and consequently contributes to enhanced hamstring injury prevention (Al Attar et al., 2017; Van Dyk et al., 2019). Since 2001, NHE training has received significant research interest as its supramaximal eccentric nature and selective activation of the knee flexors cannot be emulated by any other resistance training exercises (Brockett et al., 2001; Ebben, 2009; Timmins et al., 2015). Most athletes—especially if not familiarised—do not possess the strength capacities to perform the NHE across the entire range of motion while maintaining a constant knee extension velocity and accurate hip control. An uncontrolled NHE execution in the second

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half of the movement might diminish or even prevent neuromuscular adaptations at comparably long hamstring muscle length (Alt et al., 2018). Different research groups have demonstrated that only ~50% of unassisted NHE's range of motion (ROM) were performed in a controlled fashion (Alt et al., 2018; Delahunt et al., 2016; Ditroilo et al., 2013; Marshall et al., 2015). During the second half of the exercise, hamstring activation significantly declined and the knee angular velocity continually increased from the angle at downward acceleration (DWA) (Delahunt et al., 2016; Ditroilo et al., 2013; Marshall et al., 2015). These findings emphasised that the physical demands of a single unassisted NHE repetition are too high for most athletes (Marshall et al., 2015). It is suggested that an assisted NHE execution—particularly in the presence of insufficient eccentric knee flexor strength—might be advantageous to guide the execution of NHEs in a controlled fashion and to ensure a consistently high hamstring activation throughout the entire movement (Alt et al., 2018; Burrows et al., 2020; Matthews et al., 2015). Nonetheless, the evidence-based preventive effect of unassisted NHE training is undisputed (Mjolsnes et al., 2004; Van Dyk et al., 2019), whereas injury prevention studies implementing assisted NHEs are lacking.

Toigo and Boutellier (2006) introduced new fundamental resistance exercise determinants related to molecular and cellular muscle adaptations. They highlighted that muscular adaptations are dependent upon the range of motion, the time under tension (TUT), as well as the fractional and temporal distribution of one repetition. These parameters are aggregated in the length-at-use and contraction-mode hypotheses, meaning that muscles adapt according to their common tasks and imposed demands (Herzog et al., 1991; Savelberg & Meijer, 2003). Concerning NHE training, the fundamental resistance exercise determinants to ensure a controlled exercise execution should be the following (Alt et al., 2018; Delahunt et al., 2016; Ditroilo et al., 2013; Marshall et al., 2015):

- knee range of motion: ~90° reaching best possible knee extension (~20° to 0°),
- time under tension: ~6–7 s lasting eccentric-only movement,
- fractional and temporal distribution: aiming at an equal distribution by maintaining a constant angular velocity across the complete range of motion (100% ROM_{DWA}).

The current literature reveals a substantial lack of information about the resistance exercise determinants of NHE training. To our knowledge, three cross-sectional studies (Ditroilo et al., 2013; Marshall et al., 2015; Sarabon et al., 2019) and two longitudinal studies (Alt et al., 2018; Delahunt et al., 2016) ensured a comprehensive and precise exercise prescription by implementing kinematic analyses. Alt et al. (2018) introduced assisted 'isokinetic' (constant velocity) NHE training sessions and found altered resistance exercise determinants (e.g., lower peak moments, higher impulses, ROM_{knee} and TUT) and other accompanying parameters (e.g., higher ROM_{DWA}, lower hip flexion) compared to unassisted sessions. However, no information is available about the time course of these parameters within a NHE training session consisting of multiple sets and repetitions. Hence, there is no evidence whether an assisted or unassisted NHE should be preferentially performed and which of the different existing training regimens is advisable: 3 sets of 3 repetitions (Alt et al., 2018), 2 to 6 sets of 5 repetitions (Delahunt et al., 2016; Ditroilo et al., 2013; Marshall et al., 2015) up to 3 sets of 12 repetitions (Delahunt et al., 2016; Mjolsnes et al., 2004). This knowledge is deemed to be

essential for coaches, physiotherapists and scientists to design, perform and adjust reasonable and individual NHE training programmes for athletes of different physical performance levels and for patients in different stages of the rehabilitation process.

Thus, the aims of the present secondary analysis of an already published cohort (Alt et al., 2018) were (1) to establish the kinetic and kinematic differences between assisted and unassisted 'isokinetic' NHE execution and (2) to investigate the time course of NHE performance parameters within NHE training sessions containing 3 × 3 repetitions. Our hypotheses are as follows: (1) assisted NHE training is advantageous concerning resistance exercise determinants and (2) 3 sets of 3 'isokinetic' eccentric-only repetitions suffice to induce significant inter-repetition fatigue which impairs exercise quality.

Methods

Participants

Sixteen regional to national class male sprinters (21.6 ± 2.5 years, 181.4 ± 7.1 cm, 75.7 ± 9.8 kg) with different training history (2–17 years), training volume (4–12 h/week) and performance level (100 m season's best: 10.99–12.90 s) gave their written informed consent to voluntarily participate in the study. All of them were familiar with lower extremity resistance training, but not with NHE training. They had neither thigh muscle nor knee injuries within the last three years. The local ethics commission of the German Sport University, Cologne confirmed that the requirements of the Declaration of Helsinki were met. Within the intervention period all participants maintained their normal physical activity (e.g., sprint and resistance training) which did not demonstrably affect the relevant outcome parameters (Alt et al., 2018). Thirteen participants performed all 108 NHE repetitions (54 assisted and 54 unassisted repetitions), whereas three omitted one NHE session due to reasons which were not related to the NHE training.

Instruments

Prior to the first session, the anthropometric data of all participants were measured according to the Hanavan model (Hanavan, 1964). All NHE training sessions were executed on the isometrically operating dynamometer IsoMed2000 (D&R Ferstl GmbH, Hemau, Germany) recording raw data at 200 Hz. All relevant test parameters were synchronised and derived from the gathered moment-time curves by a self-developed software written in C++. The participants' motion was captured from above and laterally by two synchronised high-speed cameras (acA640-120gc, Basler AG, Ahrensburg, Germany) at 100 fps (TEMPLO 8.2.358, Contemplas, Kempten, Germany). Retroreflective markers (Ø = 15–25 mm) were attached on three bony landmarks of the right side of the body (acromion, trochanter major, lateral femoral epicondyle) and on a distinctive position on the lever arm representing the location of the lateral malleolus (Figure 1a, b). Two headlights (Bar Fly 200, Kino Flo® Lightning Systems, Burbank, California, USA) improved the markers' visibility. The measuring volume was statically calibrated with a rigid frame of 912 × 590 x 870 mm (L x W x H). Kinematic analyses were conducted with Vicon Peak Motus (V10.0.1, New York, New York, USA).

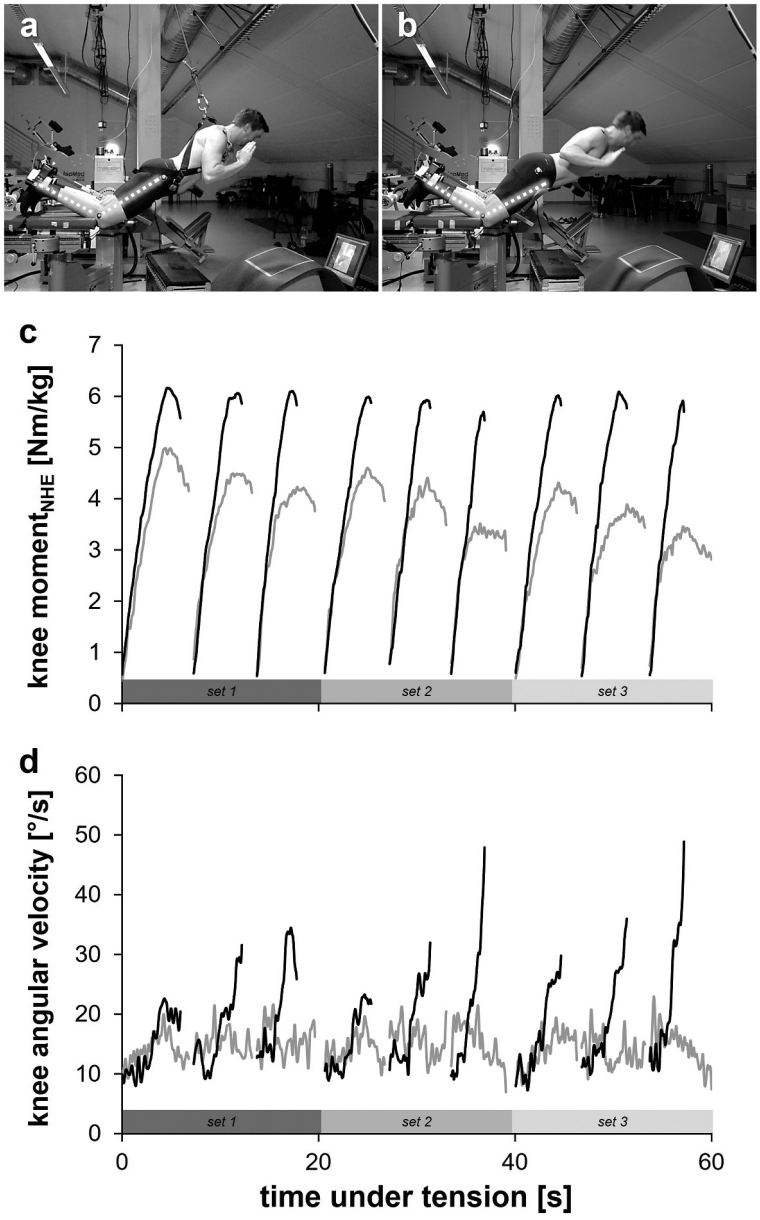

Figure 1. Representative illustrations of (a) assisted (grey lines) and (b) unassisted (black lines) NHE execution as well as their corresponding (c) moment-time and (d) angular velocity-time histories across 3 exemplary sets with 3 repetitions each.

Testing procedure

Data were derived from an earlier study (Alt et al., 2018) and re-analysed to investigate the kinetic and kinematic differences of assisted and unassisted NHEs. Prior to the analysed sessions, each participant performed six NHE training sessions (3 sessions per week with a 47 h inter-session rest), whereas the gradually alternating sequence of

assisted (A) and unassisted (U) sessions was stratified (week 1: A A A; week 2: A U A). The analysed sessions were the first two training sessions of week 3 (U A U) so that a sufficient familiarisation to NHE execution was ensured by the preceding six accommodation visits (Alt et al., 2018). Each session adhered to the identical procedure. After determining their mass, participants performed an individual warm-up (jogging, dynamic stretching) for ten minutes. The general warm-up was followed by a specific preparation at a doorway pull-up bar (Denqbar DQ-0161, Pirna, Germany) (2 x 5 repetitions, 3 min rest). These eccentric-only repetitions were executed at submaximal intensity (~60%) with a slow self-selected velocity until the DWA angle was reached. The return into the starting position was assisted by the examiner to avoid any muscular activity. Afterwards, 3 sets of 3 'isokinetic' NHEs each were performed on the dynamometer (Figure 1a, b). The inter-repetition and inter-set rest periods were set at 10 seconds and 5 minutes, respectively. Each repetition was performed solely eccentrically while the return to the starting position was to be executed without any hamstring muscle activation by a push of the arms combined with knee and hip flexion (Alt et al., 2018).

All participants kneeled on the lounger with the knee axis' orientation matching the dynamometer's axis. Prior to each repetition both heels were pushed against two pads attached to the lever arm (Figure 1a, b). The participants were asked to meet the following execution quality criteria: (1) to execute eccentric NHEs with a constant knee extension velocity of ~15°/s across the largest possible ROM_{knee} (~90°) targeting a time under tension of ~6.5 seconds per repetition and (2) to realise minimal hip flexion whilst (3) the hands were positioned in front of the shoulders (Alt et al., 2018). Minimal hip flexion was striven to increase the eccentric load on the knee flexors by maximising the lever arm of the centre of mass to the knee joint. To realise the best possible 'isokinetic' NHE execution velocity, the participants received a sagittal visual live feedback provided by a webcam recording at 30 fps (C200, Logitech, Apples, Switzerland) and a stick figure moving at target speed shown on a 17" monitor (710 N, SAMSUNG, Seoul, South Korea). Assisted NHE training sets were conducted via rope-controlled resistance by the examiner which was transferred to the back part of a climbing harness (Petzl, Crolles, France) worn by the participants (Figure 1a, 2). During every single assisted NHE repetition, the examiner adapted the resistance to the climbing harness according to the actual movement speed of the participant so that the required constant knee extension velocity was optimally achieved. Resistance was increased if participants moved faster than the projected stick figure and contrariwise.

Data processing

Raw data of NHE repetitions were stored as ASCII files before a recursive 5th order Butterworth low-pass filter with a cut-off frequency of 6 Hz was applied. The beginning and the end of a NHE repetition was determined by the first derivative of the moment-time and knee flexion angle-time curves. Knee and hip joint angles were set at 0° for full extension and increased with flexion. Figure 1 illustrates representative moment-time (c) and angular velocity-time curves (d) of assisted and unassisted NHE sets. Peak moment (PM), angle of peak moment (APM), impulse (J), time under tension (TUT), range of motion of the knee joint (ROM_{knee}), range of motion to downward acceleration (ROM_{DWA}), minimal knee flexion (KF_{min}) and maximal hip flexion angles (HF_{max}) as

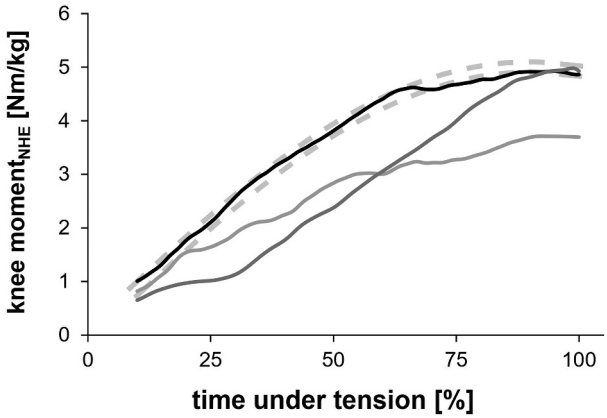

Figure 2. Knee moment curves of a single eccentric-only NHE repetition of different execution modalities: optimal unassisted execution (black), two assisted repetitions (light and dark grey) and idealistic anthropometric model (grey dashed range). The magnitude of provided rope assistance can be quantified by the area between the dashed curve and the respective grey solid line.

well as mean (ω_{mean}) and fractional knee extension velocity within 60–30° knee flexion (ω_{mean} 60–30° KF) were selected as relevant kinetic and kinematic parameters characterising NHE execution quality. The knee angle at DWA was identified as coinciding with the highest angular acceleration in the knee extension velocity-knee flexion angle curve (Delahunt et al., 2016; Ditroilo et al., 2013). At the beginning of the NHE training period, we recommended to place the knee joint in front of the shank support (Figure 1a, b) to enable that the articular cartilage of the tibia head can roll underneath the patella throughout NHE execution. However, due to inter-individual differences in perception of convenience, some participants (n = 4) chose a knee position on the lounger or on the edge of it. Misalignment of the dynamometer and the knee joint axis was mathematically corrected by calculating the resultant joint moments for each training session. Figure 2 demonstrates the agreement between resultant values and theoretical values derived from inertial segment properties based on the Hanavan model (Hanavan, 1964). All kinetic NHE parameters were normalised to body mass allowing inter- and intra-individual comparison. Percentage differences of each parameter across sets and repetitions served to estimate fatigue.

Statistical analysis

For all data, normal distribution were confirmed by the Kolmogorov-Smirnov ($\alpha \leq 0.05$). Two-way repeated measures analyses of variance identified the effects of execution mode (assisted vs. unassisted) and repetition number (repetition 1 to 9) on kinetic and kinematic NHE parameters. Bonferroni post hoc tests determined the actual p-values between the nine NHE repetitions within each execution mode. Variance homogeneity was assessed by the Levene's test to examine the between-participant variability of the two NHE execution modes. The level of significance was set at $\alpha \leq 0.05$ for all statistical tests. All statistical tests were calculated with SPSS V.23.0 (SPSS Inc., Chicago, Illinois, USA). Effect sizes are indicated as Cohen's d (≥0.8 large; 0.8–0.5 moderate, 0.5–0.2 small;

<0.2 negligible) and partial eta-squared (ηp^2) (≥0.26 large; 0.26–0.13 moderate, 0.13–0.02 small; <0.02 negligible) specifying the meaningfulness of the respective differences (Cohen, 1988).

Results

Kinetic and kinematic differences between assisted and unassisted NHE

Table 1 lists all kinetic and kinematic parameters of assisted and unassisted NHEs across 3 sets of 3 repetitions each. Both execution modes significantly differed for all analysed parameters ($p \leq 0.003$; $0.635 \leq \eta p^2 \leq 0.929$). Peak moments as well as angles of peak moment of assisted NHEs were 17% to 21% and 9° to 16° lower compared to unassisted executions. In contrast, assisted NHEs revealed significantly larger values for the impulse (+21% to +82%) (Figure 3a) and ROM_{knee} (+6° to +13°). The higher impulses were associated with a significantly greater TUT (+45% to +93%) (Figure 3b) and constantly lower ω_{mean} (−5°/s to −8°/s), whereas the greater range of motion was related to a higher range of motion at the end of the exercise (−7° to −14° KF_{min}). Due to a lower ω_{mean} 60–30° KF (−15°/s to −37°/s) (Figure 3c) the ROM_{DWA} was substantially higher (+23% to +47%) when assisted NHEs were performed (Table 1). Furthermore, assisted NHE induced a 7° to 9° smaller maximal hip flexion than the unassisted execution mode (Figure 1a, b). Between-participant variability of assisted NHEs was significantly lower for all parameters compared to unassisted NHEs, except for APM and HF_{max} (Table 1).

Time course of NHE performance parameters from 3 sets of 3 repetitions

Peak moment, impulse and time under tension showed moderate to large effects of NHE repetition in both assisted and unassisted execution mode ($p \leq 0.002$; $0.246 \leq \eta p^2 \leq 0.465$) (Table 1). Concerning the peak moment this effect was more pronounced in assisted NHEs—especially at repetition 6 and 9—, whereas the impulse (Figure 3a), time under tension (Figure 3b) as well as the fractional angular velocity within 60–30° knee flexion (Figure 3c) of unassisted NHEs were significantly reduced right away from repetition 2 onwards. The impulse of assisted NHEs decreased 11% from repetition 1 to 9 compared to an inter-repetition fatigue of 41% during unassisted execution. The same was observed for the TUT (−5% vs. −29%) and ω_{mean} 60–30° KF (+1% vs. +79%). In addition to these three parameters, ROM_{DWA} (0% vs. −31%) and ω_{mean} (+9% vs. +30%) demonstrated large interactions between mode and repetition ($p \leq 0.004$; $0.272 \leq \eta p^2 \leq 0.347$) (Table 1). Moderate interactions of mode and repetition became apparent for APM, ROM_{knee} and KF_{min} ($p \leq 0.049$; $0.131 \leq \eta p^2 \leq 0.157$). Maximal hip flexion was affected by neither repetition nor mode*repetition ($p > 0.05$; $0.056 \leq \eta p^2 \leq 0.081$).

Discussion & implications

The discussion is organised according to the aims of the study: (1) to establish the kinetic and kinematic differences between assisted and unassisted 'isokinetic' NHE execution and (2) to investigate the time course of NHE performance parameters within NHE training sessions containing 3 × 3 repetitions. The hypothesis was that assisted NHE

Table 1. Kinetic and kinematic NHE determinants (mean ± SD) obtained from 3 sets of each 3 repetitions of assisted and unassisted NHE. Significant differences ($p \leq 0.05$) between execution modes, repetitions, the interaction of both factors and variance homogeneity are highlighted. Significant differences ($p \leq 0.05$) between repetitions within one execution mode are emphasised by corresponding repetition numbers in italics.

	PM [Nm/kg]		APM [°]		J [Nm·s/kg]		TUT [s]		ROM_{knee} [°]	
	Assisted	Unassisted	Assisted	Unassisted	Assisted	Unassisted	Assisted	Unassisted	Assisted	Unassisted
Set 1 rep 1	3.61 ± 0.35	4.49 ± 0.35	38.4 ± 8.6	46.9 ± 7.2	18.1 ± 2.8	15.0 ± 5.2	6.98 ± 0.64	4.83 ± 1.17	80.2 ± 8.9	74.2 ± 7.7
rep 2	3.61 ± 0.35	4.38 ± 0.48	38.7 ± 7.1	49.6 ± 8.0	17.1 ± 2.0	13.2 ± 4.2 *1*	6.67 ± 0.49	4.56 ± 1.07 *1*	80.0 ± 7.6	73.6 ± 7.6
rep 3	3.51 ± 0.35	4.40 ± 0.41	38.0 ± 8.4	52.3 ± 9.3 *1*	16.6 ± 1.7	11.1 ± 3.3 *1*	6.62 ± 0.42	3.95 ± 0.75 *1 2*	83.1 ± 7.4	72.1 ± 8.8
Set 2 rep 4	3.62 ± 0.33	4.35 ± 0.53	37.5 ± 5.3	48.5 ± 9.1 *3*	18.0 ± 2.1 *3*	13.2 ± 5.6 *1 3*	6.96 ± 0.41 *3*	4.55 ± 1.24 *3*	80.9 ± 5.6	74.5 ± 9.9
rep 5	3.53 ± 0.32 *4*	4.37 ± 0.45	37.3 ± 8.4	52.1 ± 10.6 *1*	17.4 ± 1.8	11.8 ± 4.3 *1*	6.82 ± 0.43	4.16 ± 1.00 *1 2 4*	81.9 ± 5.8	74.5 ± 11.3
rep 6	3.45 ± 0.33 *1 2 4*	4.35 ± 0.53	38.9 ± 8.0	51.5 ± 9.9 *1*	16.7 ± 1.8 *4*	10.9 ± 3.4 *1 2 4*	6.68 ± 0.42	3.90 ± 0.76 *1 2 4*	82.4 ± 5.1	73.3 ± 9.4
Set 3 rep 7	3.59 ± 0.37 *6*	4.33 ± 0.53 *1*	36.7 ± 7.8	49.4 ± 7.0	17.9 ± 2.2 *3 6*	12.6 ± 5.3 *1 6*	6.86 ± 0.48	4.29 ± 1.18 *1 2 6*	80.9 ± 6.6	71.1 ± 7.9
rep 8	3.54 ± 0.35	4.33 ± 0.51 *1*	37.8 ± 8.7	50.9 ± 7.9 *1*	17.5 ± 1.8 *6*	10.7 ± 4.0 *1 4 7*	6.95 ± 0.41 *2*	3.80 ± 1.05 *1 2 4 5 7*	81.4 ± 5.8	72.4 ± 7.4
rep 9	3.40 ± 0.33 *1 2 4 5 7 8*	4.25 ± 0.51 *1*	36.9 ± 9.8	53.3 ± 9.2 *1 4*	16.2 ± 1.6 *1 4 5 7 8*	8.9 ± 2.6 *1 2 3 4 5 6 7 8*	6.60 ± 0.46 *4 8*	3.43 ± 0.78 *1 2 3 4 5 6 7 8*	82.0 ± 6.2	69.1 ± 10.2
Mode	$p < 0.001$; $\eta p^2 = 0.862$ *large*		$p < 0.001$; $\eta p^2 = 0.760$ *large*		$p < 0.001$; $\eta p^2 = 0.781$ *large*		$p < 0.001$; $\eta p^2 = 0.929$ *large*		$p=0.003$; $\eta p^2 = 0.681$ *large*	
Repetition	$p = 0.002$; $\eta p^2 = 0.246$ *moderate*		$p = 0.254$; $\eta p^2 = 0.083$		$p < 0.001$; $\eta p^2 = 0.465$ *large*		$p < 0.001$; $\eta p^2 = 0.391$ *large*		$p = 0.230$; $\eta p^2 = 0.082$	
Interaction	$p = 0.245$; $\eta p^2 = 0.085$		$p = 0.020$; $\eta p^2 = 0.157$ *moderate*		$p = 0.012$; $\eta p^2 = 0.204$ *moderate*		$p = 0.002$; $\eta p^2 = 0.247$ *moderate*		$p = 0.028$; $\eta p^2 = 0.141$ *moderate*	
Variance homogeneity	$p < 0.001$; $F = 13.452$		$p = 0.151$; $F = 2.072$		$p < 0.001$; $F = 66.075$		$p < 0.001$; $F = 67.280$		$p = 0.012$; $F = 6.410$	

	ROM_{DWA} [%]		KF_{min} [°]		HF_{max} [°]		ω_{mean} [°/s]		ω_{mean} 60–30° KF [°/s]	
	Assisted	Unassisted	Assisted	Unassisted	Assisted	Unassisted	Assisted	Unassisted	Assisted	Unassisted
Set 1 rep 1	100.0 ± 0.0	79.4 ± 20.5	28.7 ± 8.1	35.3 ± 7.1	8.6 ± 8.2	16.2 ± 11.0	11.4 ± 1.6	16.1 ± 4.2	11.4 ± 2.7	26.8 ± 16.3
rep 2	100.0 ± 0.0	71.6 ± 20.1 *1*	29.6 ± 8.5	36.9 ± 6.8	9.8 ± 7.7	17.4 ± 9.5	12.3 ± 1.2	16.9 ± 4.6	12.5 ± 2.1	33.2 ± 21.6 *1*
rep 3	100.0 ± 0.0	62.3 ± 27.6 *1*	27.6 ± 7.8 *2*	38.1 ± 8.2	8.8 ± 7.7	17.2 ± 7.3	13.0 ± 1.5 *1*	18.7 ± 4.3 *1 2*	13.1 ± 2.1	38.9 ± 23.7 *1*
Set 2 rep 4	100.0 ± 0.0	73.0 ± 26.4	27.3 ± 4.7	36.2 ± 9.3	8.0 ± 6.8	16.6 ± 10.6	12.1 ± 1.3 *3*	17.1 ± 4.2 *3*	11.6 ± 2.1	31.6 ± 19.9 *1 3*
rep 5	100.0 ± 0.0	69.7 ± 27.7 *1*	27.5 ± 5.3	37.0 ± 11.3	9.6 ± 8.1	17.4 ± 7.3	12.2 ± 1.5	18.5 ± 4.9 *1 4*	11.8 ± 2.1	36.8 ± 23.2 *1*
rep 6	100.0 ± 0.0	56.8 ± 24.2 *1 2 4*	27.5 ± 5.2	37.8 ± 9.6	10.4 ± 7.5	17.3 ± 8.8	12.4 ± 1.6	19.5 ± 5.2 *1 2 4*	11.8 ± 2.4	38.1 ± 23.9 *1 4*
Set 3 rep 7	100.0 ± 0.0	72.0 ± 26.5 *6*	27.5 ± 4.6	37.9 ± 8.3	8.5 ± 6.1	16.0 ± 10.2	11.9 ± 1.2 *3*	17.7 ± 5.3 *1 5 6*	11.4 ± 2.1	32.8 ± 22.1 *1 3 6*
rep 8	96.0 ± 15.9	57.1 ± 27.9 *1 2 4 7*	29.1 ± 5.1	37.9 ± 8.2	10.0 ± 7.2	17.7 ± 8.5	12.1 ± 1.3	20.0 ± 5.1 *1 2 4 7*	11.8 ± 2.1	42.3 ± 26.8 *1 4 7*
rep 9	100.0 ± 0.0	53.4 ± 26.6 *1 2 4 7*	27.0 ± 6.9	41.0 ± 10.3 *1*	9.2 ± 8.0	17.2 ± 8.0	12.5 ± 1.5	20.9 ± 5.9 *1 2 3 4 5 7*	11.6 ± 2.3	48.0 ± 29.4 *1 2 4 5 7*
Mode	$p < 0.001$; $\eta p^2 = 0.722$ *large*		$p < 0.001$; $\eta p^2 = 0.688$ *large*		$p < 0.001$; $\eta p^2 = 0.660$ *large*		$p < 0.001$; $\eta p^2 = 0.717$ *large*		$p < 0.001$; $\eta p^2 = 0.635$ *large*	
Repetition	$p = 0.004$; $\eta p^2 = 0.272$ *large*		$p = 0.273$; $\eta p^2 = 0.077$		$p = 0.282$; $\eta p^2 = 0.081$		$p < 0.001$; $\eta p^2 = 0.347$ *large*		$p = 0.006$; $\eta p^2 = 0.244$ *moderate*	
Interaction	$p = 0.013$; $\eta p^2 = 0.227$ *moderate*		$p = 0.049$; $\eta p^2 = 0.131$ *moderate*		$p = 0.482$; $\eta p^2 = 0.056$		$p = 0.020$; $\eta p^2 = 0.203$ *moderate*		$p = 0.007$; $\eta p^2 = 0.241$ *moderate*	
Variance homogeneity	$p < 0.001$; $F = 486.299$		$p < 0.001$; $F = 13.251$		$p = 0.085$; $F = 2.990$		$p < 0.001$; $F = 88.821$		$p < 0.001$; $F = 235.416$	

Footnote: PM (peak moment), APM (angle of peak moment), J (impulse (M*t)), TUT (time under tension), ROM_{knee} (range of motion of the knee joint), ROM_{DWA} (percentage of range of motion to downward acceleration in relation to ROM_{knee}), KF_{min} (mimimal knee flexion angle), HF_{max} (maximal hip flexion angle), ω_{mean} (mean angular velocity of the knee joint), ω_{mean} 60–30° KF (mean angular velocity within 60–30° knee flexion)

training is beneficial concerning resistance exercise determinants and that 3 sets of 3 'isokinetic' repetitions suffice to induce significant fatigue.

Assisted vs. unassisted NHE execution mode

The rationale for applying rope-assisted NHEs at a constant knee extension velocity was to meet the demands of the length-at-use and contraction-mode hypotheses (Herzog et al., 1991; Savelberg & Meijer, 2003). In order that muscles adapt according to their common tasks and imposed demands, NHEs should contain a high fractional and temporal portion at comparably long muscle length (Toigo & Boutellier, 2006). Adaptations at long fascicle length are of major importance because a long hamstring fascicle length has been hypothesised as an important preventive characteristic for hamstring injuries (Timmins et al., 2015). Furthermore, the slow angular velocity of 15°/s and the long concomitant time under tension of ~6–7 s complied with the recommendations for optimal muscle-tendon adaptation (Bohm et al., 2015).

As Table 1 emphasises, assisted NHE executions met the required exercise determinants, whereas unassisted NHEs revealed large between-participant variabilities and fundamental deviations of TUT, ROM_{DWA} and ω_{mean} 60–30° KF. When executing unassisted NHEs, participants were not able to maintain a constant ω_{mean} across the full ROM_{knee} (Figure 1d) (Alt et al., 2018). Right from the first repetition, significant differences between the two analysed NHE execution modes became apparent in all parameters (Table 1). With rising repetition number, the difference between assisted and unassisted NHEs gradually increased up to +82% (J) and +93% (TUT) (Figure 3a, b). These results underline that most participants did not have the strength capacities to conduct an unassisted 'isokinetic' NHE across the full ROM_{knee}. Additionally, this fact is supported by the large standard deviations of the impulse and ω_{mean} 60–30° KF (Figure 3a, c) demonstrating vast inter-individual differences in NHE execution quality. Only if these inter-individual differences are quantified, subsequent adaptations can be interpreted in a meaningful way. To our knowledge, no longitudinal study is available so far which provided this information.

Assisted NHEs revealed favourable resistance exercise determinants concerning eight out of ten analysed parameters (Table 2) with respect to ensuring high execution quality and hamstring function assessment accuracy. Maximal hip flexion cannot be explicitly assigned to one execution mode because recent studies suggested that NHE with flexed hip joints will further promote strength adaptation at long hamstring muscle length (Hegyi et al., 2019; Sarabon et al., 2019). Therefore, it is not yet clear if NHE execution with a flexed hip joint is preferable compared to best possible hip extension. Extended hip joints increase the eccentric load on the knee flexors by maximising the lever arm of the centre of mass to the knee joint. The only resistance exercise determinant which favoured unassisted NHEs was the intensity as peak moments were 17% to 21% lower during assisted executions (Table 1). Higher intensity leads to higher tendon loading (Bohm et al., 2015) and due to a higher fractional angular velocity (Figure 3c) to higher muscle-tendon strain rates. These are reasonable adaptation triggers provided that the time under tension and muscle activation are not impaired (Delahunt et al., 2016; Ditroilo et al., 2013; Marshall et al., 2015). Thus, appropriate NHE training should combine a high intensity, high impulses, as well as a high temporal and fractional distribution within a range of motion near full knee extension. If these guidelines cannot be fulfilled by unassisted NHE execution, as in the case of

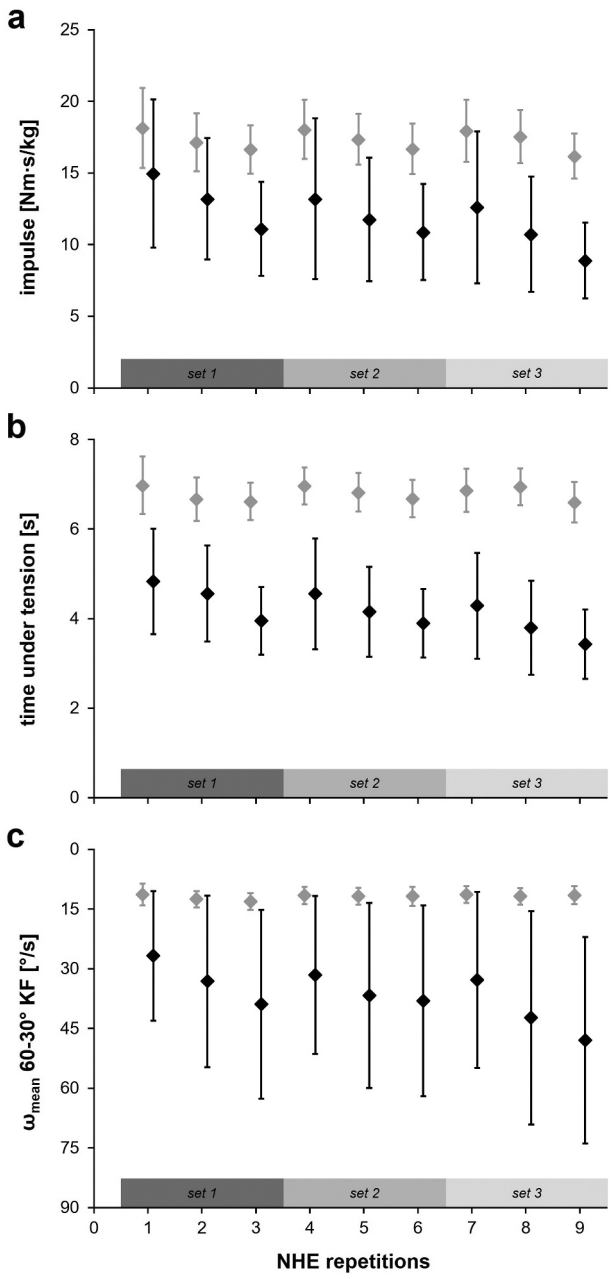

Figure 3. Inter-repetition courses of the impulse (a), the time under tension (b) and the mean angular velocity within 60–30° knee flexion (ω_{mean}) (c) obtained from 3 sets of each 3 repetitions of assisted (grey) and unassisted (black) NHE. The associated significant effects between execution mode and repetition number are indicated in Table 1.

unfamiliarised, unexperienced and weak athletes and patients, assisted NHEs should be the method of choice to execute the exercise properly and to acquire adequate strength capacities (Alt et al., 2018; Burrows et al., 2020; Matthews et al., 2015). Depending on the individual moment-angle characteristics of the athlete, assistance can be provided across

Table 2. Benefits of assisted and unassisted NHEs with respect to ensuring a high execution quality and to meeting the required exercise determinants as well as parameters which cannot be explicitly assigned.

Assisted NHE	Draw	Unassisted NHE
Angle of peak moment (APM)	Maximal hip flexion (HF_{max})	Peak moment (PM)
Impulse (J)		
Time under tension (TUT)		
Range of motion (ROM_{knee})		
Downward acceleration (ROM_{DWA})		
Minimal knee flexion (KF_{min})		
Mean angular velocity (ω_{mean})		
Fractional angular velocity (ω_{mean} 60–30° KF)		

the entire NHE repetition (Figure 2, light grey line) or within suited portions of the movement (dark grey line).

Based on the present results (Table 2), the hypothesis that assisted NHE training is advantageous concerning resistance exercise determinants can be confirmed.

Fatigue-induced effects of 3 sets of 3 NHE repetitions

Current literature introduced different NHE training regimens: 3 sets of 3 repetitions (Alt et al., 2018), 2 to 6 sets of 5 repetitions (Delahunt et al., 2016; Ditroilo et al., 2013; Marshall et al., 2015) up to 3 sets of 12 repetitions (Delahunt et al., 2016). However, there is no clear evidence how many repetitions suffice to induce a significant fatigue. Marshall et al. (2015) investigated 6 sets of 5 repetitions each conducted at an average cadence of 30°/s. They found no significant differences of fractional angular velocity, but determined reduced isokinetic peak moments (−8% to −17%) after set 2 to 6. To our knowledge, this study was the first which evaluated inter-repetition and inter-set fatigue effects by assessing NHE kinetics and kinematics.

Independent of NHE execution mode, the peak moment, impulse, time under tension, ROM_{DWA}, ω_{mean} and ω_{mean} 60–30° KF were significantly affected by repetition-induced exhaustion (Table 1). While the peak moments of unassisted NHEs remained almost unchanged, assisted NHEs suffered from a stepwise inter-repetition and inter-set decrement (Figure 1c). Unassisted NHEs revealed this phenomenon regarding inter alia the mean angular velocity (Figure 1d), impulse (Figure 3a), time under tension (Figure 3b) and the fractional angular velocity (Figure 3c). From repetition 1 to 3, the impulse (−26%), time under tension (−18%) and fractional angular velocity (+45%) of unassisted NHEs demonstrated significant changes. Although the inter-set rest of 5 minutes lead to a certain degree of recovery, these three parameters became fundamentally altered up to repetition 9 (−41%; −29%; +79%). Despite rope-assistance, peak moments, impulses and time under tension were significantly reduced comparing repetition 1 to 9 (Table 1). This effect emphasises the high physical demands of NHEs on eccentric hamstring strength (Alt et al., 2018; Brockett et al.; Ebben, 2009; Timmins et al., 2015). Other kinematic parameters, such as the knee and hip flexion angles remained nearly unchanged underlining that movement execution was not affected by fatigue.

Regarding these results, it is doubtful if exercise quality and intensity can be maintained at a sufficiently high level when performing 3 sets of 8–12 NHE repetitions per

training session. Unfortunately, no kinetic and kinematic performance parameters of such high volume NHE training regimen are available (Delahunt et al., 2016; Mjolsnes et al., 2004). However, the results of a recent study suggest that strength increases following a high volume NHE training (4 weeks of 2 x 4 x 8-10 repetitions) are not higher compared to a low volume regimen (4 weeks of 1 x 2 x 4 repetitions) (Presland et al., 2018). Furthermore, it is well-known that optimal tendon adaptation requires high intensities (~90%) and a sufficiently high time under tension (~3 s) (Bohm et al., 2015). Regarding these specific demands, low volume NHE training (2 sets of 4 reps per session) is not inferior to high volume regimen (4–5 sets of 8–10 reps) (Presland et al., 2018).

Accordingly, the hypothesis that 3 sets of 3 'isokinetic' NHE repetitions suffice to induce significant fatigue can be confirmed for assisted execution. For unassisted NHEs, already 2 sets of 3 repetitions lead to significant fatiguing effects.

Limitations & perspectives

The present results rely on a heterogenous group of sprinters varying in training history, training volume and performance level. A study incorporating homogenous samples of weak and strong athletes could identify if assisted or unassisted NHEs are more effective in promoting eccentric hamstring strength.

This study examined bilateral NHE kinetics. Prospective studies should quantify the contribution of each thigh to the generated eccentric work. Shank orientation should be modified from a negative to a positive inclination or to a horizontal orientation (Sarabon et al., 2019) to realise full knee extension (Figure 1a, b). Even during assisted NHE a minimal knee flexion of 28° remained (Table 1) due to the participants' fear of slipping off the edge. If the shank is more horizontally aligned, a greater ROM_{knee} up to almost full knee extension will be possible. Future studies should determine individual relationships between NHE training stress and NHE performance as well as related isokinetic parameters to recognise responders and non-responders. Finally, the transfer of NHE-induced improved hamstring strength to sport-specific loading and relevant performance parameters (e.g., joint power peak moments, impulses) during e.g., sprinting is of major interest.

Assisted NHE execution mode was favourable to ensure high execution quality and to meet the required exercise determinants. However, assisted NHE training requires additional equipment and is therefore less feasible for on-field training. It must still be investigated whether assisted NHE training promotes muscle-tendon adaptations at comparably long hamstring length and whether it reduces injury risk to a greater extent than unassisted NHEs. This knowledge is deemed to be essential for coaches, physiotherapists and scientists to design, perform and adjust reasonable and individual NHE training programmes for athletes of different physical performance levels and for patients in different stages of the rehabilitation process (Al Attar et al., 2017). It might contribute to a targeted injury prevention of hamstring strain injuries and related injuries such as anterior cruciate ligament tears (Van Dyk et al., 2019). Future studies should expand the previous kinetic, kinematic and electromyographic analyses to investigate the effects of different NHE training modalities (assisted vs. unassisted; bilateral vs. unilateral; neutral vs. flexed hip; unloaded vs. loaded; fast vs. slow velocity; constant-velocity vs. accelerated vs. decelerated NHEs) on eccentric hamstring strength and fascicle length

(Presland et al., 2018; Sarabon et al., 2019; Timmins et al., 2015) to develop individualised training regimen adjusted to different profiles of requirements (e.g., high vs. low physical performance level).

Practical implications

NHE execution on specific devices (Hegyi et al., 2019; Opar et al., 2013; Sarabon et al., 2019) or isometric dynamometers (Alt et al., 2018) is recommended to assess exercise determinants, performance parameters and exercise quality. If not available, wall bars, doorway pull-up bars or any other solid and rigid object can serve to provide adequate counter-pressure for the heels. Despite extensive research since 2001, standardised NHE execution is lacking as most studies execute partner-assisted NHE training (Delahunt et al., 2016; Ditroilo et al., 2013; Mjolsnes et al., 2004). This procedure has two major disadvantages that impede its scientific demands. Firstly, the applied resistance from hands to shanks/heels is mostly insufficient for mature athletes to perform a NHE across the full ROM_{knee}. Beyond inadequate strength capacities of the athletes this deficient fixation results in ROM_{DWA} of below 30% (Delahunt et al., 2016; Ditroilo et al., 2013; Marshall et al., 2015). Secondly, looking from a more scientific perspective, it is not possible to determine the generated forces/moments quantifying e.g., training stimuli and fatigue and derive adequate exercise determinants.

Assisted NHE execution is recommended to acquire proper exercise quality by lower inter-individual variability and to reduce the fear of uncontrolled falling, especially within early training stages of inexperienced athletes or patients. It is advisable to conduct NHEs always across the full ROM_{knee} close to full knee extension (~20° to 0° KF). The intensity of effort is important to increase hamstring strength and fascicle length (Roig et al., 2009; Timmins et al., 2016). That's why a nearly constant movement velocity during the second half of the NHE's ROM_{knee} (=100% ROM_{DWA}) should be particularly paramount where peak moments occur (Figures 1d, 2, 3c). In order to achieve this guideline, assistance can be provided by a partner being located in front of the athlete and adjusting the pressure of his hands to the athlete`s shoulders according to the respective movement velocity.

The use of additional weights during unassisted NHE execution with extended hip joints should be exclusively limited to experienced athletes to avoid injuries caused by excessive strain rates (Hegyi et al., 2019). Unilateral NHE training is only recommended if an uncontrolled falling is restricted by any sort of assistance. Due to increasing inter-set fatigue during unassisted NHEs which highly varied between participants (Figure 3a-c), it might be more reasonable to use muscular failure (Toigo & Boutellier, 2006)—associated with a large increase of angular velocity at the end of a repetition (Figure 1d, rep6 and rep9)—as completion criterion of a set rather than a prescribed repetition number. This can contribute to more individualised NHE training regimen respecting actual daily performance state.

Conclusion

Assisted NHEs were favourable to ensure a high execution quality to meet the required exercise determinants. Furthermore, between-participant variability was significantly lower in assisted than unassisted NHEs. Only if the athlete's/patient's physical performance level is adequate, controlled full-ROM NHEs in unassisted fashion are recommended to ensure

a consistently high hamstring activation throughout the entire movement and subsequent adaptations at comparably long hamstring muscle length. If these guidelines are fulfilled, exercise intensity can be reasonably estimated and reproduced. An exercise volume of 3 sets of 3 eccentric-only NHE repetitions induced substantial fatigue within and across sets and can be considered as adequate exercise stimulus Further research is needed to establish which NHE execution modalities elicit the best adaptations.

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